Intermediary Asset Pricing in Over-the-Counter Markets

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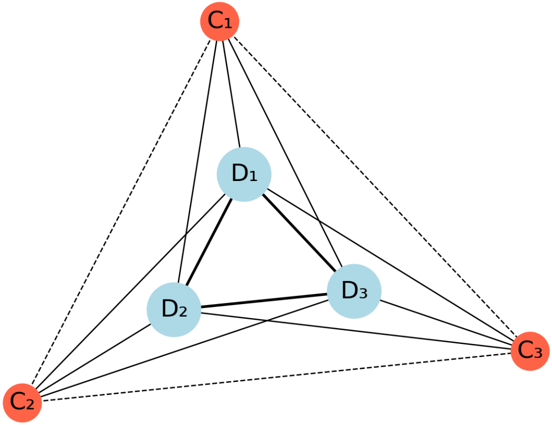
Balance Sheet Costs in OTC Markets

Dealers (D)...

- Intermediate customer (C) order flow
- Manage inventory subject to **internal and regulatory risk limits**
  - Value-at-Risk (VaR) Limits
  - Banks' Supplementary Leverage Ratio
- Trade in the interdealer market to distribute risk within the dealer sector



Intersecting OTC and Regulatory Frictions Have Been Costly

MARKETS

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Regulators face questions over how to avoid more frequent dysfunction

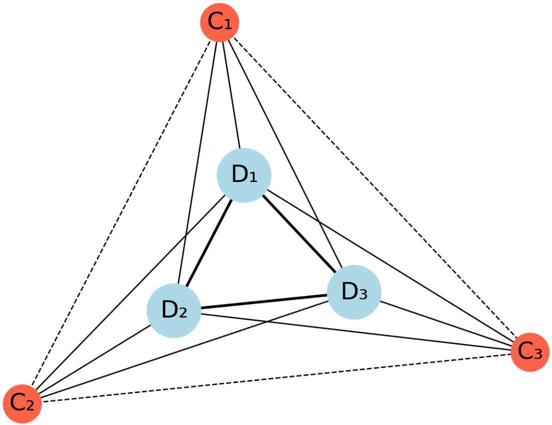
By Nick Timiraos

WSJ PRO Updated July 28, 2021 3:05 pm ET

How do dealer inventory costs affect market liquidity and asset prices?

The answer depends on:

1. Market frictions - How easy is it to find a buyer (seller)
   - Between customers
   - Between dealers
2. The distribution of dealer inventory holdings
   - How willing are different dealers to intermediate



The Average Dealer Inventory Matters for Prices

Intermediary Asset Pricing
He and Krishnamurthy (2013), Adrian, Etula, and Muir (2014), He, Kelly, and Manela (2017), He, Khorrami, and Song (2022)

- Average capital ratios of dealers price risky assets

Over-the-Counter Search
Duffie, Gârleanu, and Pedersen (2005), Üslü (2019)

- With quadratic utility only average dealer inventory matters for prices

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Our contribution:
Present a model where the entire distribution of dealer inventories determines asset prices in order to explain the effect of balance sheet constraints in the Basel III era.

Theoretical Model

- Time is continuous, $t \in [0, \infty)$
- A measure 1 of dealers trade in a frictional market
- Random matching of dealers with each other and their short lived customers
- Dealers hold a_t units of the asset and maximize lifetime expected utility

$$\mathbb{E}_t \int_0^\infty e^{-rt} u(a_t) dt$$

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- **Key assumption:** If $u'' < 0$ and $u''' > 0$
  - Dealer risk-aversion depends on the level of their inventory
  - i.e. Curvature is endogenous to their intermediation activity
  - Dealers at different points of the inventory distribution will behave differently

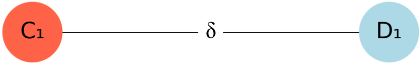
Theoretical Model

Trade

<u>With customers</u>

A dealer receives orders from their customers at rate δ and receive θ of the joint surplus from trade, S_c.

Customers have exogenous valuations of the asset $\epsilon \sim G(\epsilon)$.



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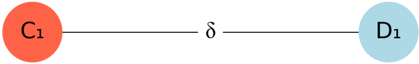
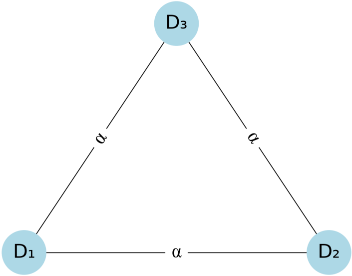


With other dealers

Dealers trade with other dealers at rate α and split the joint surplus, S_d, evenly.



Equilibrium

A fixed point between F and V

$$rV(a;F) = \underbrace{u(a)}_{\text{Flow utility}} + \underbrace{\delta\theta \int S_c(a,\varepsilon)dG(\varepsilon)}_{\text{Surplus from customer trade}} + \underbrace{\frac{\alpha}{2}\int S_d(a,\tilde{a})dF(\tilde{a})}_{\text{Surplus from dealer trade}}$$

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<u>After Trade Positions</u>

<u>With customers</u>

$$a = V'^{-1}(\epsilon)$$

<u>With other dealers</u>

$$\bar{a} = \frac{a+\tilde{a}}{2}$$

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$$\dot{F}(a) = \underbrace{\delta(1 - G(V'(a)) - F(a))}_{\text{Flow from customer trade}} + \underbrace{\alpha\left(\int\int \mathbb{I}\left(\frac{a_1+a_2}{2} \leq a\right) dF(a_1)dF(a_2) - F(a)\right)}_{\text{Flow from dealer trade}}$$

Implications

With $u(a) = -e^{-\gamma a} + \kappa_0 a - \kappa_1 a^2$

With increased curvature, dealers...

- Hold larger, less dispersed, inventories
- Trade less and at worse prices

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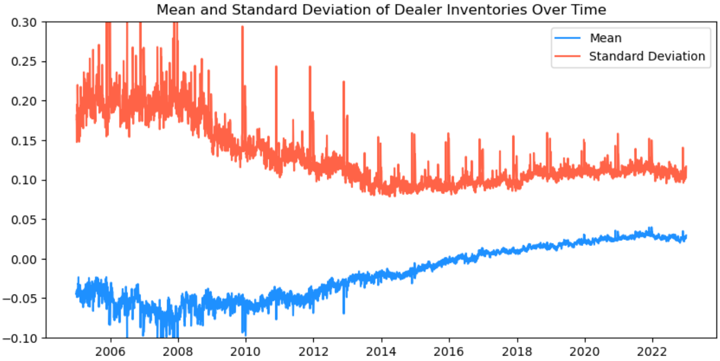
Data

Academic TRACE

- Every corporate bond transaction since 2002
  - CUSIP, Price, quantity, transaction time
  - Anonymized dealer identifiers linked to each transaction
- Reconstruct dealer inventories across time



Note: Data shows mean and standard deviation of largest 50 dealers scaled by outstanding.
Source: FINRA Academic TRACE, Mergent FISD.

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The Inventory Distribution Matters More in Riskier Markets

$$\Delta CreditSpread_t^i = a + \beta \Delta \mu_t^{Inv} + \eta \Delta \sigma_t^{Inv} + \boldsymbol{\theta' \Delta Controls_{i,t}} + \epsilon_t^i$$

Table: Changes in IG and HY Credit Spreads

	IG	HY	All
$\Delta \mu_t^{Inv}$	-0.860***	-2.926***	-0.926***
	(-14.21)	(-2.69)	(-12.90)
$\Delta \sigma_t^{Inv}$	1.929***	6.636***	2.292***
	(27.64)	(8.71)	(24.27)
Mean adj. R^2	0.338	0.418	0.345
Median adj. R^2	0.356	0.426	0.361
Obs.	343,826	34,754	382,507
Bonds	4,804	579	5,281

Next Steps: Identification and Estimation

Identification

- Meeting rates, α and δ: Number of dealer-to-customer and interdealer transactions each month
- Dealer utility parameters, γ and κ: Transaction volumes and prices
- Distribution of customer valuation shocks, $G(\epsilon)$: Price dispersion of dealer-to-customer transactions

Estimation

- For CARA utility, we have a closed form solution and can estimate via GMM

Counterfactuals

- Decreased regulations (Basel III Endgame), decreased γ or κ
- Targeted asset purchases, increased α or δ by sub-market

Conclusion

- The relevance of balance sheet costs for pricing and liquidity depends on market frictions and the distribution of dealer inventories
- Our model explains the post-crisis decrease in the dispersion of dealer inventories through increased balance sheet costs
- Fluctuations in the distribution of dealer inventories explain changes in bond credit spreads